# Md Mahbub Hasan Bappy

Microbiologist
Food QA/QC & Product Development




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## CORE SKILLS

- Microbiological testing: endotoxin (BET/LAL), sterility, bioburden, microbial limit
- Environmental monitoring and water tests
- GMP / ISO documentation and traceability
- Audits and staff training
- Food law and labelling (EU)
- HACCP, IFS, BRCGS, ISO 9001 (knowledge from studies)
- Food product development, sensory evaluation
- PCR / RT-PCR, ELISA, MIC, plating and staining

## SOFTWARE

- MS Office (Word, Excel, PowerPoint)
- SPSS, XLSTAT, Python, R / PAST
- unipark (surveys), RedJade (sensory)
- PRODI, EndNote

## LANGUAGES

English: fluent
German: B2 (University of Kassel,

## PROFILE

Microbiologist (M.Sc. Microbiology) with one year of hands-on quality assurance experience in a regulated medical-device plant, including GMP/ISO documentation and participation in ISO, regulatory and internal audits. Currently completing the joint M.Sc. International Food Business and Consumer Studies (University of Kassel and Hochschule Fulda) with knowledge of food quality management, EU food law, product development and sensory testing. Structured, analytical and detail-oriented. Available immediately: full-time, part-time, working student, internship or master's thesis in a company.

## WORK EXPERIENCE

### Executive Officer (Microbiologist), Quality Control / Quality Assurance

JMI Hospital Requisite Mfg. Ltd., Munshiganj, Bangladesh | 10/2022 – 10/2023

- Ran microbiological release testing on 2–5 batches a day (about 4,000 units each): endotoxin, sterility, bioburden, biological and chemical indicators, environmental monitoring, water tests, autoclave checks
- Kept test reports, logbooks and traceability records in line with GMP and ISO
- Took part in ISO, regulatory and internal audits (all passed); contributed to ISO 14001 documentation
- Core person responsible for setting up a new laboratory, including calibration and machine checks; introduced new test methods
- Trained junior colleagues, new staff and lab assistants

### Research Assistant, COVID-19 Detection Lab

Noakhali Science and Technology University (NSTU), Bangladesh | 10/2021 – 12/2021

- RNA extraction, master-mix preparation, PCR / RT-PCR handling and result interpretation
- Media preparation, culture, biochemical tests and pathogen isolation

### Research Volunteer, NSTU Research Cell

Noakhali, Bangladesh | 02/2020 – 02/2021

- Data and sample collection, surveys, analysis and event management

## EDUCATION

### M.Sc. International Food Business and Consumer Studies

University of Kassel and Hochschule Fulda | 10/2024 – present | taught in English

- 84 ECTS completed; top grades (1.3) in Food Product Development, Food Processing, Digital Transformation in the Food Sector; 1.7 in Quality Management in the Food Business and Innovative Product Development
- Master's thesis (30 ECTS) still open; looking for a company-based thesis

### M.Sc. Microbiology (CGPA 3.63 / 4.00, 2nd of 7)

Noakhali Science and Technology University, Bangladesh | 2022 – 2023 | taught in English

### B.Sc. (Hons) Microbiology (CGPA 3.69 / 4.00)

Noakhali Science and Technology University, Bangladesh | 2016 – 2021 | taught in English

## AWARDS

- National Science and Technology (NST) Fellowship 2022–23, Bangladesh Ministry of Science and Technology
- NSTU merit scholarship 2017
- Government merit scholarships

## TRAINING

- WHO: Basic Microbiology; COVID-19 detection, prevention and control (2021)
- IEDCR Bangladesh: Real-time PCR detection of COVID-19 (2021)
- German B2 certificate, University of Kassel (2026)

## MEMBERSHIPS

- Graduate Microbiologist Society (since 2020)
- NSTU Science Club, Biofilm Microbiology Club

## REFERENCES

Available on request.

## PROJECTS

### Applied Research Project (18 ECTS, grade 1.3): AI in Idea Generation for Seafood Alternative Products

04/2025 – 03/2026 | University of Kassel and Hochschule Fulda; supervisor Prof. Dr. A. Quadt; team of five

- Compared AI-, consumer- and expert-generated vegan seafood ideas: 12 concepts, online survey with 73 valid responses
- Statistical analysis with XLSTAT and Python (ANOVA, t-tests, correlation, post-hoc tests); AI ideas scored highest on innovativeness, consumer ideas on likeability and willingness to buy

### Innovative Product Development: "Tomato Air Crunch Ball" (grade 1.7)

10/2025 – 03/2026 | Hochschule Fulda with Antonius Hof Fulda; team of three

- Own product idea: fried chickpea snack with 6.6% tomato powder; lab trials, formulation, PRODI nutrition calculation and allergen identification
- Wrote the results and discussion chapter: process flow with frying as critical control point, hedonic and just-about-right consumer test (20 valid responses), penalty analysis

### Food Product Development: muesli bar for children (grade 1.3)

2025 | Antonius Hof Fulda; team of four

- 12 ideas generated, top 5 shortlisted, final concept chosen by about 30 classmates; lab prototypes with raspberry-powder coating, PRODI nutrition calculation, allergen check
- Sensory test with 25 participants (hedonic and just-about-right scales)

### M.Sc. project: MRSA from avian eggs (NSTU)

Master's project work | Noakhali Science and Technology University

- Screening of 100 duck, quail and poultry-farm eggs for *S. aureus* and methicillin resistance: enrichment, Baird-Parker agar, Gram stain, disc diffusion (13 resistant, 16 intermediate isolates)

### Course papers

- Digital systems for food-fraud prevention (olive oil adulteration case; blockchain, IoT, traceability, EU Regulation 2017/625)
- CSR strategy of General Mills along the Cheerios oat value chain (ISO 26000, CSRD, SDGs)
- Plant-based dairy alternatives: consumer trends and market potential

### B.Sc. projects

- Bacteria in minimally treated fruits and vegetables (8 samples); antibiotic-resistant bacteria in campus drinking water; in-silico characterisation of Alexandrium fundyense proteins